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3 May 1966

MILITARY SPECIFICATION

PRESSURE TRANSDUCER

(For Torpedo MK46 Mod 0 and Mod 1)

1. SCOPE

1.1 This specification covers requirements for the procurement of one type of pressure transducer (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-P-116	Preservation, Method of
MIL-P-18177	Plastic sheet, Laminated, Thermosetting, Glass Fiber Base, Epoxy-Resin
MIL-R-5757	Relays, Electrical (Excluding Thermal for Electronic and Communication-Type Equipment), General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals
MS33586	Metals, Definition of Dissimilar

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DRAWINGS

Bureau of Naval Weapons

2539249	Pressure Transducer, Specification Control Drawing

OTHER PUBLICATIONS

ANA Bulletin No. 147	Specifications and Standards of Non-Government organizations released for Flight Vehicle Construction
Federal Test Method Std 151 Method 812	Metals, Test Methods Salt Spray and Humidity Tests
NAS 710	Resistors: Variable, Precision

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall be provided and shall:

(a) Be manufactured according to procedures and requirements and using tooling intended for the production lot. A complete set of manufacturing drawings and procedures used to fabricate the preproduction samples shall be supplied to the cognizant naval activity in a sealed package to provide assurance that unauthorized changes to the product are not effected. Should it be considered necessary to open this package, it shall be in the presence of the supplier. Upon satisfactory completion or termination of the contract the design data shall be returned to the supplier.

(b) Consist of three (3) pressure transducers.

(c) Be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in section 2, unless specified in the contract or order (see 6.2).

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3.2.2 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Standard MIL-STD 143 and ANA Bulletin No. 147.

3.2.3 Inspection and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities and provide personnel for performing all acceptance tests specified herein. The test equipment shall be adequate to permit performance of the required acceptance tests for the pressure transducer.

3.2.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in Standard MS33586.

3.2.5 Soldering. All solder connections shall be made in accordance with MIL-STD-440.

3.2.6 Fungus proof material. Materials that are nutrients for fungi shall not be used unless necessary. Where used and not hermetically sealed they shall be treated with a fungicidal agent.

3.3 Performance and product characteristics.

3.3.1 Physical characteristics. The pressure transducer shall conform to the physical requirements delineated on BuWeps Drawing 2539249.

3.3.1.1 Using hydraulic oil per Specification MIL-H-5606, the change in volume required to pressurize the unit from 0 to 750 psig shall not exceed 0.030 cubic inches.

3.3.2 Pressure medium. The pressure transducer shall meet all requirements specified herein using any pressure medium as follows:

(a) Hydraulic Oil per Specification MIL-H-5606.

(b) Sea water at temperatures above 27 degrees F.

(c) Dry nitrogen.

3.3.3 Electrical requirements. All requirements are to be met the first time a measurement is made. Cycling to condition the unit prior to a test is not acceptable.

3.3.3.1 Potentiometer. The potentiometer shall:

(a) Meet all requirements with 23 to 33 volts direct current (vdc) or volts root-mean-square (vrms) at 4800 ±100 cycles per second (cps) steady state over a temperature range of 0 to 168 degrees F.

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(b) Be capable of withstanding 55 vrms at 4800 ±100 cps for 5 seconds over a temperature range of 0 to 168 degrees F each time the unit is energized.

(c) Be wirewound type.

3.3.3.1.2 DC resistance, terminal 1 to terminal 3, (see figure 1) shall be 5000 ±250 ohms at 79 degrees F. Temperature coefficient shall be ±0.035 percent per degree F maximum.

3.3.3.1.3 Impedance between terminal 1 and terminal 3 shall be 5000 ±300 ohms at 26 ±2 vrms, 4800 ±100 cps.

3.3.3.1.4 Resolution shall be 0.0 to 1.0 pound per square inch (psi) over a temperature range of 0 to 168 degrees F. The potentiometer shall be wound with 500 turns minimum over its active range.

3.3.3.1.5 Repeatability including incremental hysteresis when cycling over any 20-30 psi range shall not exceed 2.0 psi over a temperature range of 0 to 168 degrees F. This requirement includes resolution and all other conditions with no vibration and the case temperature held constant within ±5 degrees F.

3.3.3.1.6 Wiper shall withstand 10 ma dc or rms continuously and meet requirements of 3.3 after removal of the current.

3.3.3.1.7 Noise generated during operation of the potentiometer shall not exceed 300 ohms equivalent noise resistance (see 6.3.14). This requirement shall apply with the following simultaneous conditions or any combination thereof:

(a) Pressure rate of 0.0 to 30.0 psi per second.

(b) Wiper current of 100 ±10 microamperes (ua) dc.

(c) Noise spectrum; low pass with 50 ±5 kilocycles (kc) cutoff with a nominal slope of 6 db per octave beyond 50 kc.

(d) Vibration as described in operating vibration schedule of table I.

(e) Temperature range of 0 to 168 degrees F.

3.3.3.1.8 Total error in ratio of output voltage (terminals 1 and 2) to applied input voltage (terminals 1 and 3) shall not exceed limits specified in figure 1. This requirement shall apply with the following conditions:

(a) Load (terminal 1 to 2); greater than 9 megohms.

(b) Errors due to nonlinearity, hysteresis, resolution or any other condition except test equipment.

(c) Environmental conditions as follows.

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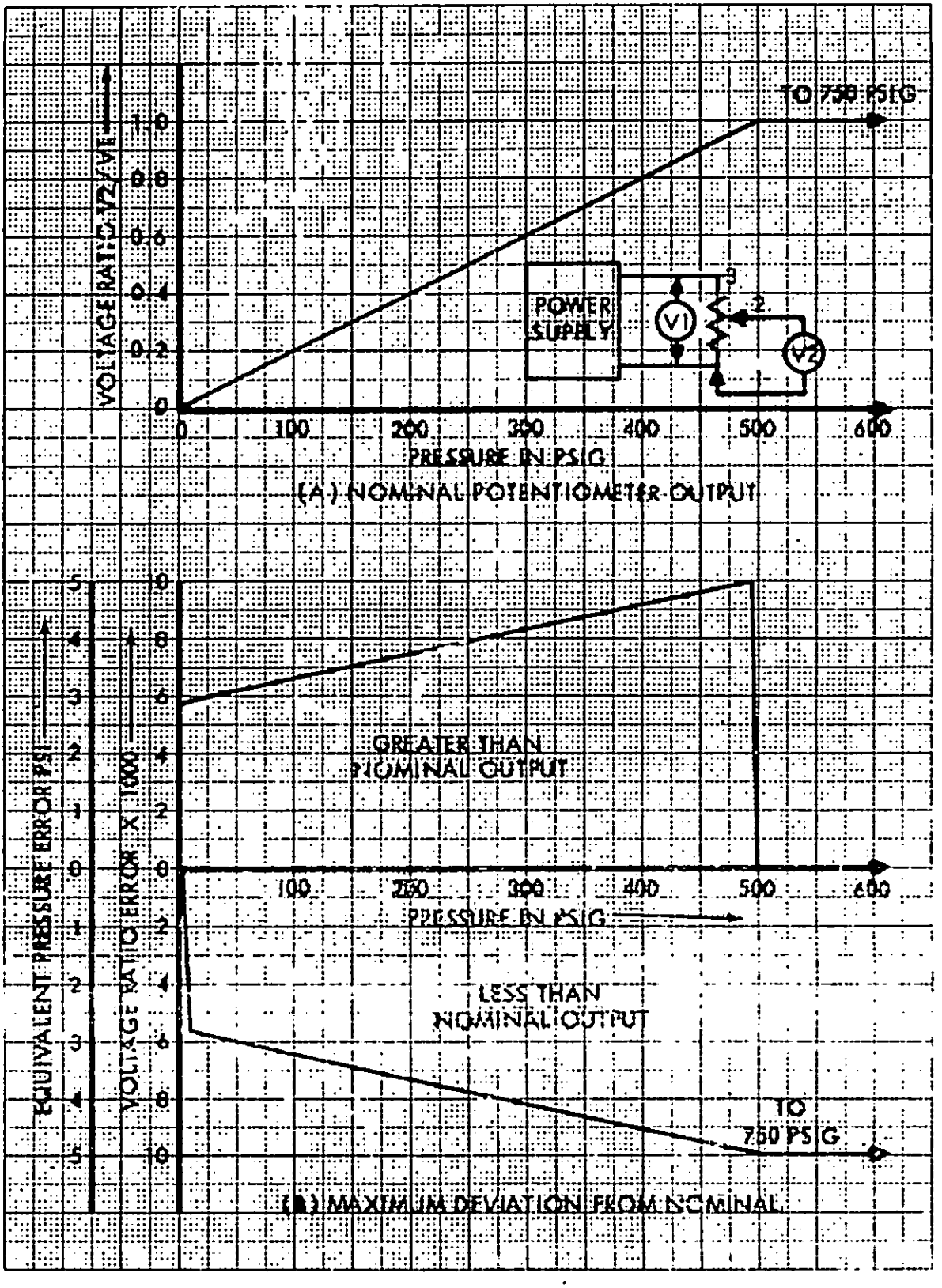


Figure 1. Potentiometer Characteristics

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1.	Temperature;	65 to 80 degrees F
2.	Vibration;	none
3.	Altitude;	Normal ground atmospheric pressure 24 to 32 inches mercury
4.	Humidity;	Room ambient, up to 95 percent

(d) Applies for increasing pressure and at 0 psig and 500 psig.

3.3.3.1.9 The transducer shall not require more than a 2.5 psi adjustment in pressure from the pressure at 79 ±3 degrees F required to obtain any voltage ratio when tested in accordance with 4.7. This applies at any temperature in the range from 0 to 168 degrees F other than specified in 3.3.3.1.8(c)1.

3.3.3.2 Switches.

3.3.3.2.1 Each switch shall close (see 6.3.8) within the specified range as pressure increases from zero psig or any pressure less than the specified pressure and shall remain closed for pressures greater than the specified range. Each switch shall open (see 6.3.9) within the specified range as pressure decreases from 750 psig or any pressure above the specified range to 750 psig and shall remain open for pressure less than the specified range including errors due to any condition except test equipment.

3.3.3.2.2 Operating (specified) range at the environmental condition of 3.3.3.1.8 shall be:

S1: 8.9 to 12.5 psig
S2: 22.3 to 25.8 psig
S3: 42.7 to 46.3 psig

3.3.3.2.3 Operating (specified) range at any other temperature in the range from 0 to 168 degrees F shall be:

S1: 7.5 to 14.0 psig
S2: 21.0 to 27.5 psig
S3: 41.0 to 48.0 psig

3.3.3.2.4 Switch contact resistance shall not exceed 10.0 ohms at 3.0 ±0.2 ma dc at any pressure greater than the pressure specified in 3.3.3.2.2 and 3.3.3.2.3.

3.3.3.2.5 Each switch shall meet all requirements when switching 5.0 ±0.2 ma dc resistive load at 30 ±2 vdc.

3.3.3.2.6 With decreasing pressure the difference between the pressure when switch S3 opens and the pressure required for a potentiometer voltage ratio equivalent to 55.5 ±0.1 psig at 79 ±3 degrees F shall be 9.0 psi minimum, over the ambient temperature range of 0 degrees to 168 degrees F.

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3.3.3.3 Response time. The pressure transducer shall meet requirements of 3.3.2 through 3.3.3.2.6 while being subjected to a pressure rate of change of 30 ±3 psi per second over a temperature range of 0 degrees to 168 degrees F.

3.3.3.4 Hi-pot. There shall not be more than 100μa leakage current when 500 vdc is applied for 10 to 15 seconds between isolated circuits (see 6.3.17) or all electrical circuits and the mounting surfaces. Leakage current between any individual terminal and the mounting surfaces shall not exceed 25μa.

3.3.4 Life. Life requirements shall be as specified for the following unit conditions (see 6.3.1).

3.3.4.1 Storage life, nonoperating. The pressure transducer, preserved and packaged, is intended to have an unattended storage life of 5 years.

3.3.4.2 Storage life, ready or ready nonoperating. The pressure transducer mounted in service installation, is intended to have an unattended storage life of one year.

3.3.4.3 Operating life.

3.3.4.3.1 Full pressure cycles. The pressure transducer shall meet the requirements of 3.3.2 through 3.3.3.3 for 400 cycles of 0 to 500 to 0 psig and followed by 50 full pressure cycles (6.3.12). With 50 additional full pressure cycles the pressure transducer shall meet all requirements of paragraphs 3.3.2 through 3.3.3.3 except the upper and lower tolerances shall be widened by 1 psi.

3.3.4.3.2 Incremental pressure cycles. The pressure transducer shall meet the requirements of 3.3.2 and 3.3.3 after 60 energized potentiometer incremental pressure cycles (see 6.3.15) at any pressure, and 15 energized switch incremental pressure cycles (see 6.3.16) for each switch.

3.4 Environmental requirements. Unless otherwise specified, the pressure transducer shall be capable of conforming to the performance and product characteristics requirements of this specification after being exposed, at any temperature in the applicable range, to the environments under all unit conditions specified in this section. Refer to figure 2 for axes of the pressure transducer.

3.4.1 Vibration. The pressure transducer shall be capable of withstanding the vibrations shown in table I in the three axes shown in figure 2.

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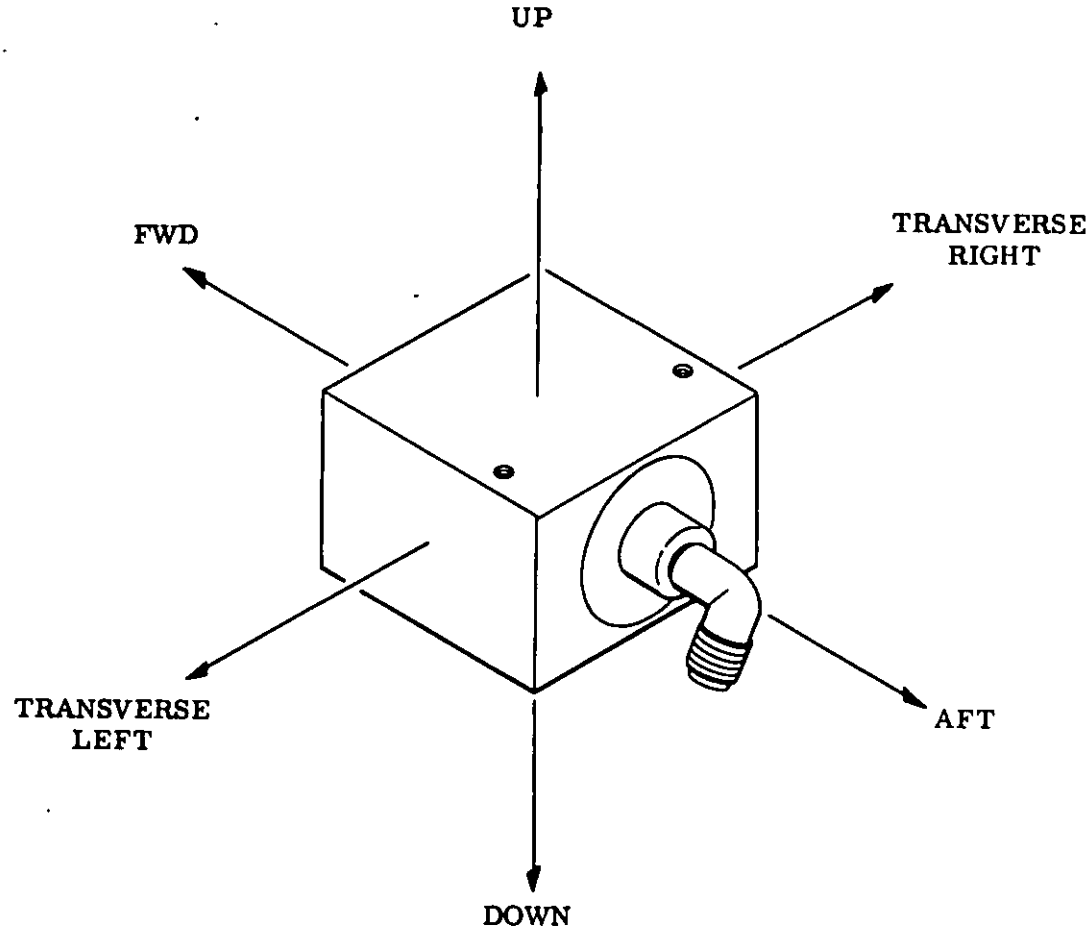


Figure 2. Axes of the Pressure Transducer

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Table I

Vibration Schedule

Condition	Double Amplitude (in) minimum	Acceleration (g) minimum	Frequency (cps)
Nonoperating	--	1.0	5-60
Ready nonoperating; Ready (Energized per 4.5.2.8)	0.2 -- 0.036 --	-- 3.0 -- 10.0	5-17 17-40 40-74 74-500
Operating (Energized per 4.5.2.8)	0.050 -- 0.020 --	-- 1.0 -- 3.0	5-20 20-32 32-56 56-500

3.4.2 Shock. The packaged pressure transducer (see 5.1.1) shall be capable of withstanding the shock of being dropped on a nonyielding surface from a height of 36 inches. Additionally, the pressure transducer, in the condition shown in table II, shall withstand shocks specified in table II.

Table II

Shock Schedule

Condition (See 6.3.1)	Peak Magnitude (g's) ±10%	Duration (milliseconds)	Acceleration Direction*	Number of Impacts
Ready nonoperating	60	8.0 to 10.0	3 fwd or aft axis 3 left or right axis 3 up or down axis	9
Ready	45	18.0 to 20.0	Long. fwd axis	3
	85	0.5 to 1.5	3 up or down axis 3 right or left axis	6
	300	0.5 to 1.5	Long. aft axis	3

*See figure 2 for axes of pressure transducer

3.4.3 Acceleration. The pressure transducer shall be capable of withstanding accelerations specified in table III.

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Table III

Acceleration Schedule

Condition	Magnitude (g's) ±10%	Acceleration Direction
Ready	25	Longitudinal Fwd
Ready nonoperating	35	Longitudinal Aft

3.4.4 Temperature. The pressure transducer shall be capable of withstanding temperatures for condition specified.

3.4.4.1 Nonoperating and ready nonoperating. The pressure transducer shall be capable of withstanding ambient temperatures between -45 and +140 degrees F.

3.4.4.2 Ready and operating. The pressure transducer, when operated in an environment with a temperature between 0 degree F and 168 degrees F, shall meet performance and product characteristics of this specification after exposures to temperatures between zero degree F and +140 degrees F.

3.4.5 Pressure.

3.4.5.1 The pressure transducer shall be capable of withstanding pressures from 2.7 to 20.0 psia externally with or without the operating medium being applied.

3.4.5.2 The pressure transducer shall be capable of withstanding a vacuum of 26 to 28 inches of mercury (Hg) through the pressure fitting for thirty minutes minimum with normal ground atmospheric pressure, 24 to 32 inches Hg, externally.

3.4.5.3 The pressure transducer shall be capable of withstanding a burst pressure (see 6.3.10) of 1000 psig.

3.4.6 Humidity and salt laden atmosphere. The pressure transducer shall withstand exposure to salt laden atmosphere such as encountered in a shore installation.

3.4.7 Salt water. The unit shall show no deterioration in performance or physical integrity as the result of salt water being used as a pressure media.

3.5 Workmanship. Workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the pressure transducer shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative

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of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the pressure transducer shall be classified as follows:

(a) Preproduction inspection.

1. Preproduction sample (see 4.3.1)
2. Preproduction test (see 4.5)

(b) Periodic production inspection.

1. Periodic production sampling (see 4.3.2)
2. Periodic production tests (see 4.6)

(c) Acceptance inspection.

1. Acceptance sampling (see 4.3.3)
2. Acceptance tests and examinations (see 4.7)

4.3 Sampling.

4.3.1 Preproduction sample. A preproduction sample of three pressure transducers, manufactured in accordance with 3.1 shall be subjected to preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the pressure transducer by the supplier, prior to approval of the preproduction sample, shall be at supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from pressure transducers that have satisfactorily passed quality conformance inspections of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production, randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

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(b) Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively, starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit (see 6.3.18) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Acceptance sampling. Tests and examinations specified in 4.7 shall be conducted on each pressure transducer to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the pressure transducer shall be subjected to tests under following conditions:

(a)	Temperature	Room ambient (65 to 80 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure
(d)	Vibrations	None (unless otherwise specified)
(e)	Switch Load	5.0 ±0.2 ma resistive at 30 ±2 vdc
(f)	Potentiometer	30 ±2 vdc constant voltage source between terminals 1 and 3

4.5 Preproduction tests. The pressure transducer within preproduction sample (identified as pressure transducer 1, 2, and 3) shall be subjected to tests specified in table IV at test conditions of 4.4 unless otherwise specified. Certified inspection data covering results of preproduction tests shall be forwarded to the activity specified in the contract or order. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting the tests.

(b) Run or test plan used.

(c) Serial number of the pressure transducer.

(d) Test results of preproduction tests outlined in table IV.

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Table IV

Preproduction Test Schedule

Sequence	Test Description	Pressure Transducer Number		
		1	2	3
1	Acceptance Tests 4.7 and test to requirements of 3.3.3.1.3, 3.3.3.1.4, 3.3.3.3 and 3.3.3.4	x	x	x
2	Vibration (4.5.2.1)	x		x
3	Shock (4.5.2.2)	x	x	
4	Acceleration (4.5.2.3)	x		x
5	Temperature, storage (4.5.2.4.1)	x	x	
6	Temperature, operating (4.5.2.4.2)	x	x	
7	External pressure test (4.5.2.5)	x	x	
8	Vacuum test (4.5.2.6)	x	x	
9	Humidity and salt laden atmosphere test (4.5.2.7)	x		x
10	Salt water test (4.5.2.9)	x	x	
11	Life test (4.5.2.8)	x		x
12	Acceptance Tests 4.7* and test to requirements of 3.3.3.1.3, 3.3.3.1.4, 3.3.3.3, and 3.3.3.4	x	x	x
13	Burst pressure (4.5.2.10)	x	x	x

*For units which were subjected to life tests see 3.3.4.3.1 and 4.5.2.8.2.

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform preproduction tests specified herein:

4.5.1.1 A vibration table capable of vibrating a 1 pound load, plus fixture weight at 10 g's maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitude of 0.20 inch.

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4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 20 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g magnitude for period of one minute.

4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of -45 to +168 degrees F.

4.5.1.5 A chamber capable of providing the environmental conditions specified in 4.5.2.7.

4.5.1.6 Acceptance test equipment specified in 4.7.2.

4.5.1.7 Test equipment to measure equivalent noise resistance (see 6.3.14). Suggested test method is per NAS 710.

4.5.1.8 Vacuum equipment capable of providing a vacuum of 26 to 28 inches of Hg.

4.5.1.9 A Megohmmeter capable of supplying a potential of 200 to 500 vdc and measuring leakage resistance of 1 megohm or greater of the circuit under test.

4.5.1.10 Hydraulic pressure source capable of generating pressures specified in 4.5.2.8.

4.5.1.11 Test equipment to measure switch chatter (see 6.3.13). Suggested test method is in MIL-R-5757D Amendment 3.

4.5.2 Detailed preproduction tests.

4.5.2.1 Vibration.

4.5.2.1.1 Nonoperating. The packaged pressure transducer (see 5.1.1) shall be placed on a vibration platform, fenced on all sides so that the pack cannot tip over or fall off the platform. With the packaged pressure transducer in the vertical position, gradually increase the frequency from 5 cps until the pack just begins to leave the platform. At this point, run test for 15 minutes. Repeat procedure with the packaged pressure transducer laying on its side.

4.5.2.1.2 Ready nonoperating and ready. The unpackaged pressure transducer, mounted on a support frame, shall be rigidly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in table I to the pressure transducer along the axes shown in figure 2. Vibration will be conducted at three temperature conditions; the pressure transducer stabilized at room temperature, at -45 ±5 degrees F and +140 ±5 degrees F. The resonant modes in each of the three axes, at each of the three temperatures, shall be established by energizing the switches (see 4.5.2.8) and applying a voltage of 32 ±2 vdc to terminals 1 and 3 of the potentiometer with a load resistor (RN60C4642F) across terminals 1 and 2. The pressure shall be zero (0). The

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frequency of applied vibration shall be slowly varied from 5 cps to 500 cps to 5 cps as specified in table I. Resonant modes shall be detected by switch chatter or potentiometer wiper movement. If more than four resonances are encountered for any one temperature condition on any one axis, the four most severe resonance points shall be chosen for the tests at resonance, for that one axis and that one temperature condition. The total vibration time in each axis shall be 180 minutes with 60 minutes of this time with transducer stabilized at -45 ±5 degrees F; 60 minutes with pressure transducer stabilized at +140 ±5 degrees F and 60 minutes at room ambient temperature. The pressure transducer shall be vibrated for 10 minutes at each resonance point for each of the specified temperatures in each axis; the remainder of the 60 minutes shall be used in cycling per table V. The frequency rate during cycling shall be logarithmic such that 1/12 of the total cycling time is required to proceed from the lower to higher and back to the lower extremity of frequency range.

Table V

Resonant and Cycling Schedule

No. of Resonances	Time at resonance at each temperature in each axis, and at each resonance (Min)	Cycling time at each temperature in each axis (Minutes)
0	0	60
1	10	50
2	10	40
3	10	30
4	10	20

4.5.2.1.3 Operating Vibration. The pressure transducer shall be mounted in a support frame and oriented in accordance with figure 2. Subject the unit to the vibration frequencies and amplitudes for operating condition of table I along each of its three mutually perpendicular axes. The switches shall be energized (see 4.5.2.8) and a voltage of 30 ±2 vdc shall be applied at terminals 1 and 3 of the potentiometer with a load resistor (RN60C4642F) across terminals 1 and 2. The unit shall be subjected to a log frequency sweep at each pressure of 8.0 ±0.2, 13.5 ±0.2, 21.0 ±0.2, 27.0 ±0.2, 41.5 ±0.2, 47.5 ±0.2, 250 ±5, and 450 ±5 psig. The frequency sweep rate shall be logarithmic such that 15 minutes is required to proceed from the lower to higher and back to the lower extremity of frequency range. The switches and potentiometer wiper shall be monitored at all times. There shall be no switch chatter (see 6.3.13) at any time. The variation in potentiometer wiper voltage at any fixed input pressure shall not exceed 150 mv peak-to-peak.

4.5.2.2 Shock

4.5.2.2.1 Nonoperating. The packaged pressure transducer (see 5.1.1) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact

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surface which shall be essentially nonyielding stone, concrete or similar material. The container, with the packaged unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3 and 6-1-4, and side 1 and face 5 as defined in figure 3.

4.5.2.2.2 Ready nonoperating and ready. Mount the pressure transducer onto a shock test fixture so that a minimum of 90 percent of table input of the fixture will be applied to the pressure transducer under test. Subject the pressure transducer to shocks listed in table II, in conditions specified. The duration shall be measured by an accelerometer mounted on fixture to indicate applied shock.

4.5.2.3 Acceleration. The pressure transducer shall be subjected to acceleration magnitudes and directions required by table III for one minute at room temperature for each condition specified.

4.5.2.4 Temperature.

4.5.2.4.1 Temperature, ready nonoperating. The unpackaged pressure transducer shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

(a) Heat chamber to +140 ±5 degrees F in 2.5 ±0.5 hours.

(b) Maintain at +140 ±5 degrees F for a minimum of 4.0 hours, then within 2.5 ±0.5 hours cool chamber to -45 ±5 degrees F.

(c) Maintain at -45 ±5 degrees F for 4 hours minimum, then within 2.5 ±0.5 hours return chamber to room temperature.

repeat this schedule for a minimum of three cycles.

4.5.2.4.2 Temperature, operating. The pressure transducer, mounted in accordance with figure 4, shall be tested in accordance with 4.7 and shall meet requirements of 3.3.3.1.9, 3.3.3.2.3, and 3.3.3.2.6 under the following conditions:

(a) The pressure transducer shall be conditioned for one hour at a temperature of +140 to +171 degrees F, then transferred to an environment at +168, ±3 degrees F and tested.

(b) The pressure transducer shall be conditioned for one hour at a temperature of zero ±3 degrees F, and tested.

4.5.2.5 External pressure. Subject the pressure transducer to atmospheric pressures of 2.7 and 20.0 psia for periods of 30 minutes minimum at each pressure with pressure port sealed.

4.5.2.6 Vacuum. With the case at normal ground atmospheric pressure, subject the pressure transducer to a vacuum of 26 to 28 inches of Hg applied at the pressure port for 5 minutes minimum.

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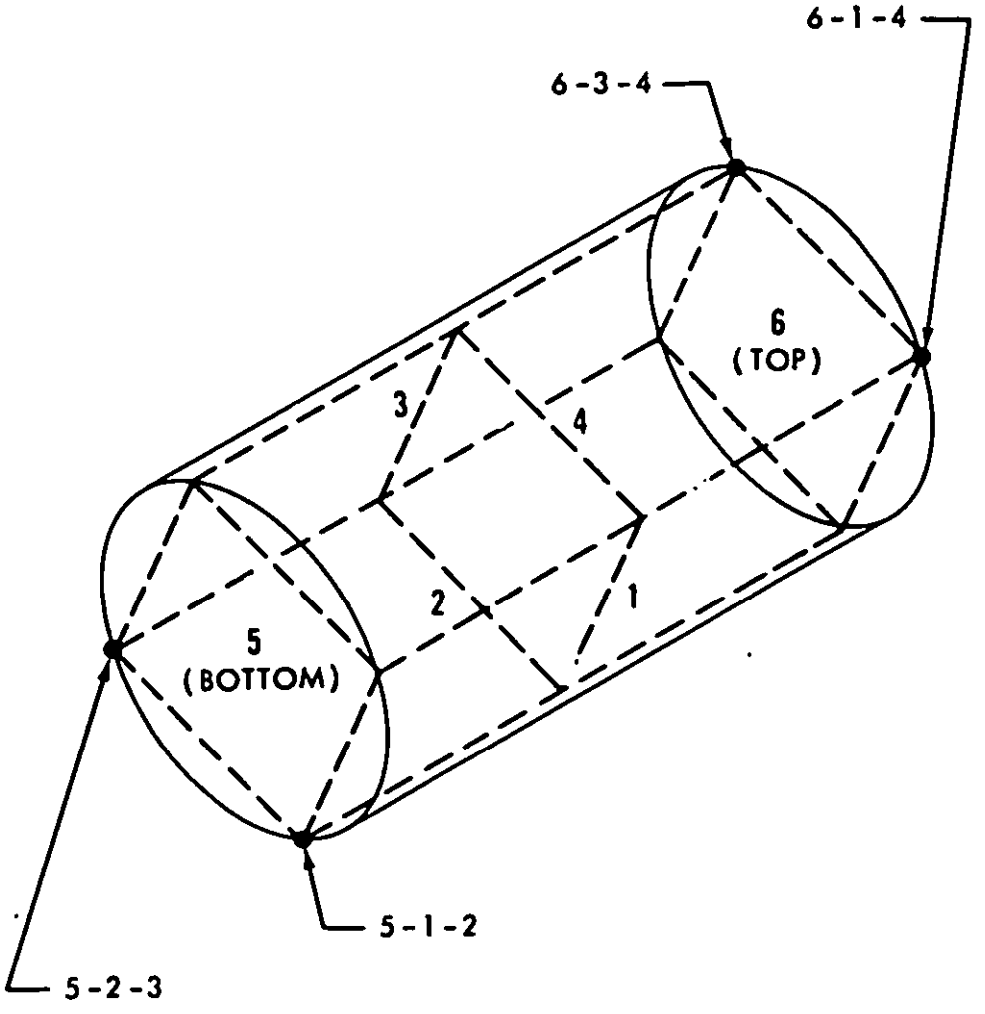


Figure 3. Definition of Faces and Corners of Shipping Container

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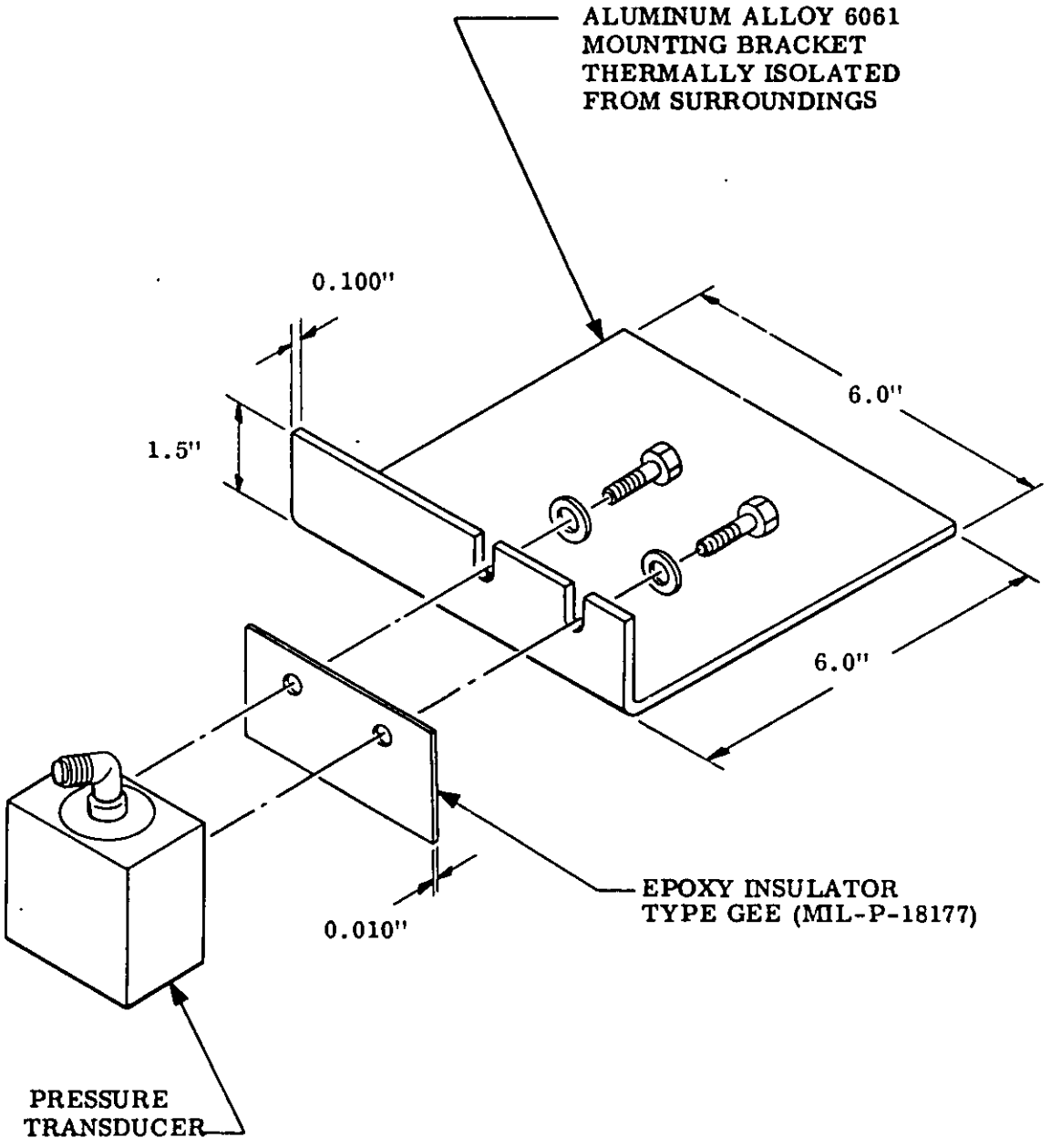


Figure 4. Thermal Mounting for Temperature Test

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4.5.2.7 Humidity and salt laden atmosphere. Subject the pressure transducer to a synthetic sea-water spray test per Federal Test Method Std 151, Method 812, for a period of eight hours minimum. During the eight hours, vary the temperature between 40 degrees F and 90 degrees F every 2 hours. During the test the pressure port shall be sealed. After exposure, the unit shall remain at room environment for 24 hours. The unit shall not be washed or dried. Subject the pressure transducer to a Hi-Pot test of 200 vdc between all terminals and case. Leakage resistance shall be not less than 1 megohm. Wash with fresh water and dry unit for 24 hours at room ambient conditions. Unit shall then meet requirements of 3.3.3.4.

4.5.2.8 Operating life. The pressure transducer shall be energized with a 100 ±10μa dc constant current source between terminals 1 and 2 and a 5.0 ±0.2 ma resistive load at 30 ±2 vdc load for each switch and a pressure medium of hydraulic oil and subjected to the following operating life tests.

4.5.2.8.1 Incremental pressure operating life.

4.5.2.8.1.1 The pressure transducer shall be operated for 15 switch incremental pressure cycles for each switch at an average pressure where the switch will open and close during the peak-to-peak pressure variation. One cycle shall be a pressure variation of 4 to 6 psi peak-to-peak at 3 ±1 cps for 7.0 ±0.2 minutes. The pressure variation versus time may represent a sine wave to a square wave.

4.5.2.8.1.2 The pressure transducer shall be operated for 15 potentiometer incremental pressure cycles at an average pressure of 250 ±5 psig. One cycle shall be a pressure variation of 4 to 6 psi peak-to-peak at 3 ±1 cps for 7.0 ±0.2 minutes. The pressure variation versus time may represent a sine wave to a square wave.

4.5.2.8.2 Full pressure operating life. The pressure transducer shall be operated for 400 cycles of 0 to 500 to 0 psig, and then for 50 cycles of 0 to 750 to 0 psig. During each cycle the pressure must remain at maximum pressure for 5 seconds minimum, the rate of pressure change shall be 30 +0 -15 psi/sec. After each 50 cycles the pressure transducer must be tested in accordance with the provisions of 4.7. After completion of the 450 cycles the pressure transducer shall be operated for an additional 50 cycles of 0 to 750 to 0 psig and then tested in accordance with the provisions of 4.7, except the pressure tolerance limits on the switches be widened 1 psi and the pot error band shall be widened 0.02 turns at each limit.

4.5.2.9 Salt water. Introduce through the pressure transducer's pressure inlet, salt water solution to completely fill the pressure chamber. After the pressure chamber has been kept filled for 48 hours minimum, empty the salt water solution from the pressure chamber. After standing in room environment for 24 hours minimum, the pressure chamber shall be washed out. The salt water solution shall be 5 percent sodium chloride prepared as in Federal Test Method Standard No. 151, Method 811, paragraph 2.1.4.

4.5.2.10 Burst pressure. The pressure transducer shall be subjected to a pressure of 1000 to 1050 psig for 10 minutes minimum. The unit shall show no leakage during this test. Other requirements are waived after this test.

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4.5.3 Preproduction acceptance criteria. Acceptance shall be based on satisfactory completion of test sequence items 1 through 13 of table IV as applicable.

4.6 Periodic production test. Periodic production sample, selected in accordance with 4.3.2, shall be subjected to tests of 4.5 as modified by table VI.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for the conduction of all tests of Sections 4.5 and 4.7.

4.6.2 Periodic production test schedule. The periodic test sample shall be subjected to the tests noted in table VI in the order specified.

4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2.1 and 4.7.1.3.

4.7 Acceptance tests and examinations. Unless otherwise specified, each pressure transducer delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions.

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in order shown.

4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in Reading column, indicating that requirements were satisfactorily met.

4.7.1.3 Acceptance criteria. Failure of the pressure transducer to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.7.1.4 Acceptance tolerances. The tolerances in the following procedure reflect the test equipment and operator accuracy. The intent is to procure pressure transducers that meet paragraph 3.3 herein. Therefore, if more accurate test equipment (such as a ratiometer or dead weight tester) is used, the tolerances must be adjusted to reflect this accuracy of measurement. The procuring activity reserves the right to perform such a test and show conformance to 3.3.

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Table VI

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1, 4, Etc.	Sample No. 2, 5, Etc.	Sample No. 3, 6, Etc.
Vibration (4.5.2.1.1, 4.5.2.1.2)	1	1	
Operating Vibration (4.5.2.1.3)	2	2	2
Shock (4.5.2.2)	4		1
Acceleration (4.5.2.3)	3		
Temperature Ready Non-operating (4.5.2.4.1)	5	3	
Hi-Temperature Operating (4.5.2.4.2,a)	13	5	
Lo-Temperature Operating (4.5.2.4.2,b)	11		6
External Pressure (4.5.2.5)	12	4	
Vacuum (4.5.2.6)	10		3
Humidity and salt laden atmosphere (4.5.2.7)	6		4
Hi-pot (3.3.3.4)	7-9		5
Life (4.5.2.8)	14		
Salt water (4.5.2.9)	8		
Burst Pressure (4.5.2.10)	16	7	
Acceptance tests of 4.7* and test to requirements of 3.3.3.1.4, 3.3.3.3	15	6	7

*For units which were subjected to life tests see 3.3.4.3.1 and 4.5.2.8.2

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4.7.3 Acceptance test equipment. Items of test equipment shown in table VII are required to perform acceptance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

Table VII

Test Equipment

Item	Identification
Power Supply 0 to 30 VDC, 225 ma max	Hewlett-Packard, Model 721A
Digital DC Voltmeter (Range: 0.001 to 1.000 volts dc; Accuracy: 0.1% of reading ±1 digit; Input impedance: 10.2 megohms)	Hewlett-Packard, Model 3440A with preamplifier 3441A
Resistance Bridge (Resistance: 0.001 to 10 megohms; Accuracy: ±1%)	General Radio, Model 1650 A
Hydrostatic Pressure Simulator (0 to 750 psig)	See figure 9
Pressure Transducer Test Sets	See figures 6 and 8
Oscilloscope: (Sweep range: 20 usec/cm to 5 ms/cm; Frequency range: DC to 100 kc min.) Vertical sensitivity: 3 cm or 1 inch/30 mv input)	
Low Pass Filter: (Cutoff frequency: 50 kc; Average slope: 5 to 7 db octave	
Vibration Table: (Capable of providing inputs of table I for operating vibration)	

NOTE

A ratiometer may be used for testing when available. If a ratiometer is used, delete references to Heli-pot adjustment for null. The NULL control dial reading will now be read directly by the ratiometer with the decimal point moved one place to the left. (See 4.7.1.4.)

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4.7.3 Electrical tests.

NOTE

Tests shall be conducted under the following conditions:

Load (terminal 1 to 2); greater than 9 megohms

Environmental conditions:

Temperature;	65 to 80 degrees F
Vibration;	None, except as noted
Altitude;	Normal ground atmospheric pressure
Humidity;	Room ambient, up to 95 percent

4.7.3.1 Initial tests.

WARNING

At no time should any potential be applied between terminals 1 and 2 or 3 and 2, which can produce a current greater than 2 ma.

4.7.3.1.1 Using the resistance bridge, measure the dc resistance of the pressure transducer potentiometer (terminal 1 to terminal 3). The resistance shall be 4700 to 5300 ohms at 79° F. (Temperature coefficient ±0.035 percent per degree F maximum.)

4.7.3.1.2 Connect the multimeter to the following terminals and the resistance shall be greater than 1 megohm for each reading.

(a) Case to each terminal

(b) Terminal 1 to 4, 5, 6, 7, 8 and 9

(c) Terminal 4 to 6, 7, 8 and 9

(d) Terminal 6 to 8 and 9

4.7.3.2 Noise test.

4.7.3.2.1 Mount the pressure transducer on the vibration table, oriented in the Up and Down axis as shown in figure 5.

4.7.3.2.2 Connect the pressure transducer for test as shown in figures 6 and 7. Make sure that the pressure fitting to the pressure transducer is secure.

4.7.3.2.3 Set the power supply to ON and adjust for a 30.0 ±0.1 vdc output. Record value.

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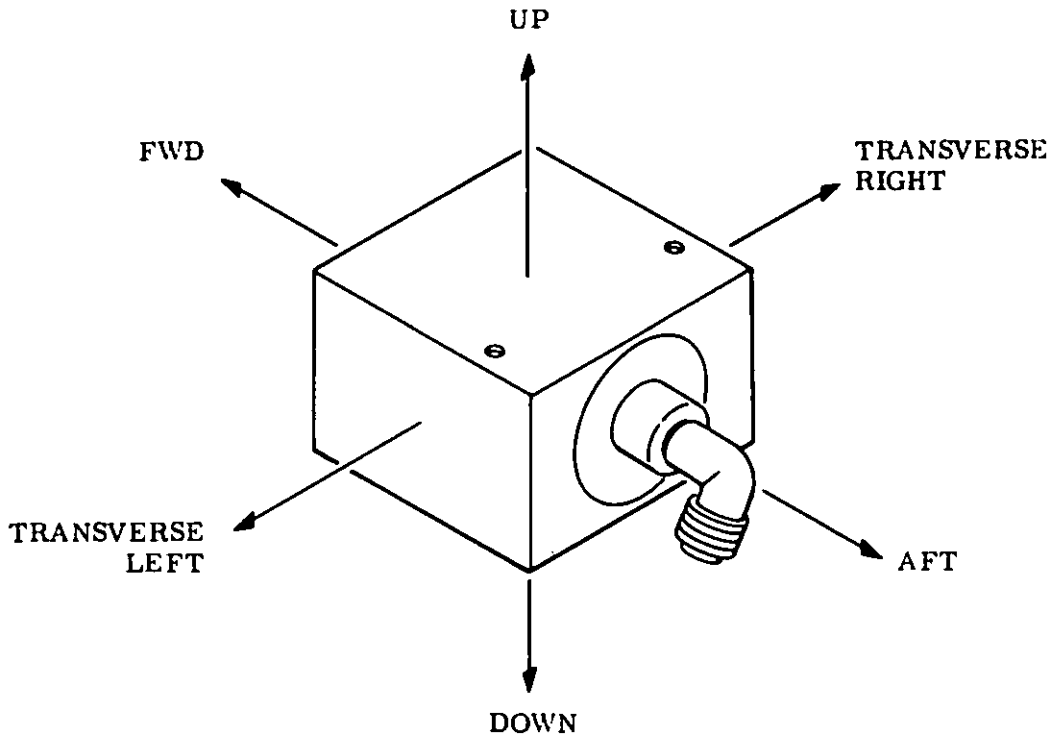


Figure 5. Axes of Vibration

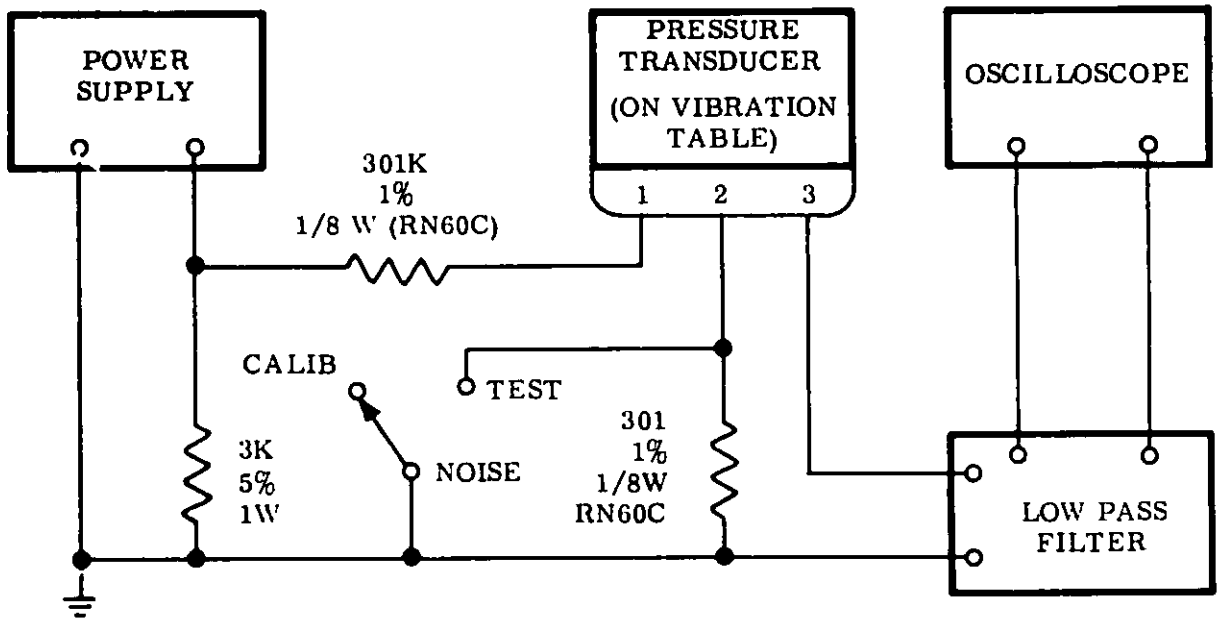


Figure 6. Noise Test Circuit

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1. TANK DRY NITROGEN
2. PRESSURE GAGE, 0-4000 PSI (US GAGE 22660-1)
3. PRESSURE REGULATOR 0-750 PSIG (VICTOR SR-10)
4. PRESSURE GAGE,*0-1000 PSIG (US GAGE 17965-1)
5. PRESSURE REGULATOR (GROVE MODEL 15-SX)
6. NEEDLE VALVE CUTOFF (GROVE SERIES T2)
7. CUTOFF VALVE, 15 to 250 PSIG (REPUBLIC 680-1)
8. PRESSURE GAGE,**0-60 PSIG (HEISE BOURDON TYPE)
9. CUTOFF VALVE, 15 to 250 PSI (REPUBLIC 680-1)
10. PRESSURE GAGE,** 0-250 PSIG (HEISE BOURDON TYPE)
11. CUTOFF VALVE 250-1000 (REPUBLIC 680-2)
12. PRESSURE GAGE,** 0-550 PSIG (HEISE BOURDON TYPE)
13. PRESSURE TRANSDUCER

*Gage 4 accuracy to be ±0.5% full scale range, with 5 psig increments maximum.

**Gage accuracy to be ±0.1% full scale range. Gages 10 and 12 increments to be 0.2% full scale maximum. Gage 8 increments shall be 0.2 psig maximum.

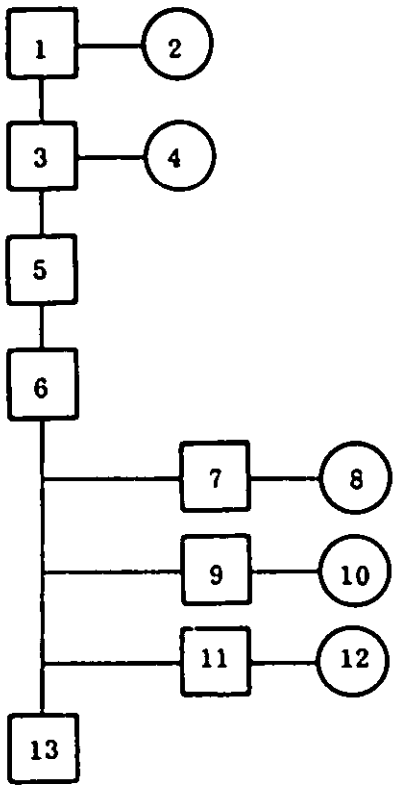


Figure 7. Test Connections for Hydrostatic Pressure Simulator

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4.7.3.2.4 Set the NOISE switch to TEST. Set the oscilloscope to ON. (Allow sufficient time for warmup to prevent reference level shifts.) Set the scope trace to the bottom line of the graticule. Set the NOISE switch to CALIB. Adjust the vertical amplifier gain for full scale deflection. This is the 300 ohm noise calibration level.

4.7.3.2.5 Set the NOISE switch to TEST. Apply power to the vibration table at levels described in table I, operating condition. Vary the vibration frequency from 5 to 500 cps using a sweep rate which requires 5 minutes minimum for the excursion. Concurrent with the vibration, the input pressure to the pressure transducer shall be cycled from 0 to 500 to 0 psig at a rate of 15 to 30 psig/sec. The pressure cycling shall continue for the duration of the vibration sweep.

4.7.3.2.6 If any deflection of the oscilloscope trace beyond the 300 ohm noise calibration level occurs, except at the zero and 500 psig pressure reversal points, the pressure transducer shall be rejected. Turn off power to vibration table.

CAUTION

Care must be taken that extraneous inputs to the test equipment used in figure 6, such as AC power line transients, do not cause false noise indications.

4.7.3.2.7 Repeat 4.7.3.2.1 through 4.7.3.2.6 for the transverse and FWD-AFT axes of figure 5.

4.7.3.2.8 Disconnect test circuits and remove the pressure transducer from the vibration table.

4.7.3.3 Operational tests.

4.7.3.3.1 Connect pressure transducer in the test set (shown in figure 8). Use the test setup as shown in figure 9. Do not use chassis ground on the digital dc voltmeter.

4.7.3.3.2 Make sure the pressure line fitting to the pressure transducer is secure.

4.7.3.3.3 Adjust the regulator knob on the Hydrostatic Pressure Simulator, hereinafter called HPS, to maximum counterclockwise (CCW) position.

4.7.3.3.4 Adjust the Leak Check cut-off valve on the HPS to its maximum CCW position. Assure the gages are indicating 0 psig.

4.7.3.3.5 Turn Power Supply on. Position the pressure transducer test set POWER switch to ON. Position the METER switch to VDC. Adjust the NULL control dial to 0.00. Set digital DC voltmeter for maximum sampling rate. Adjust the power supply for 30.0 ±0.1 vdc indication on the digital DC voltmeter (hereinafter referred to as DVM). Record value.

4.7.3.3.6 Position the METER switch to POT. The DVM shall indicate 0.175 VDC maximum.

4.7.3.3.7 Position the METER switch to 100'. The DVM shall indicate 29.6 VDC minimum.

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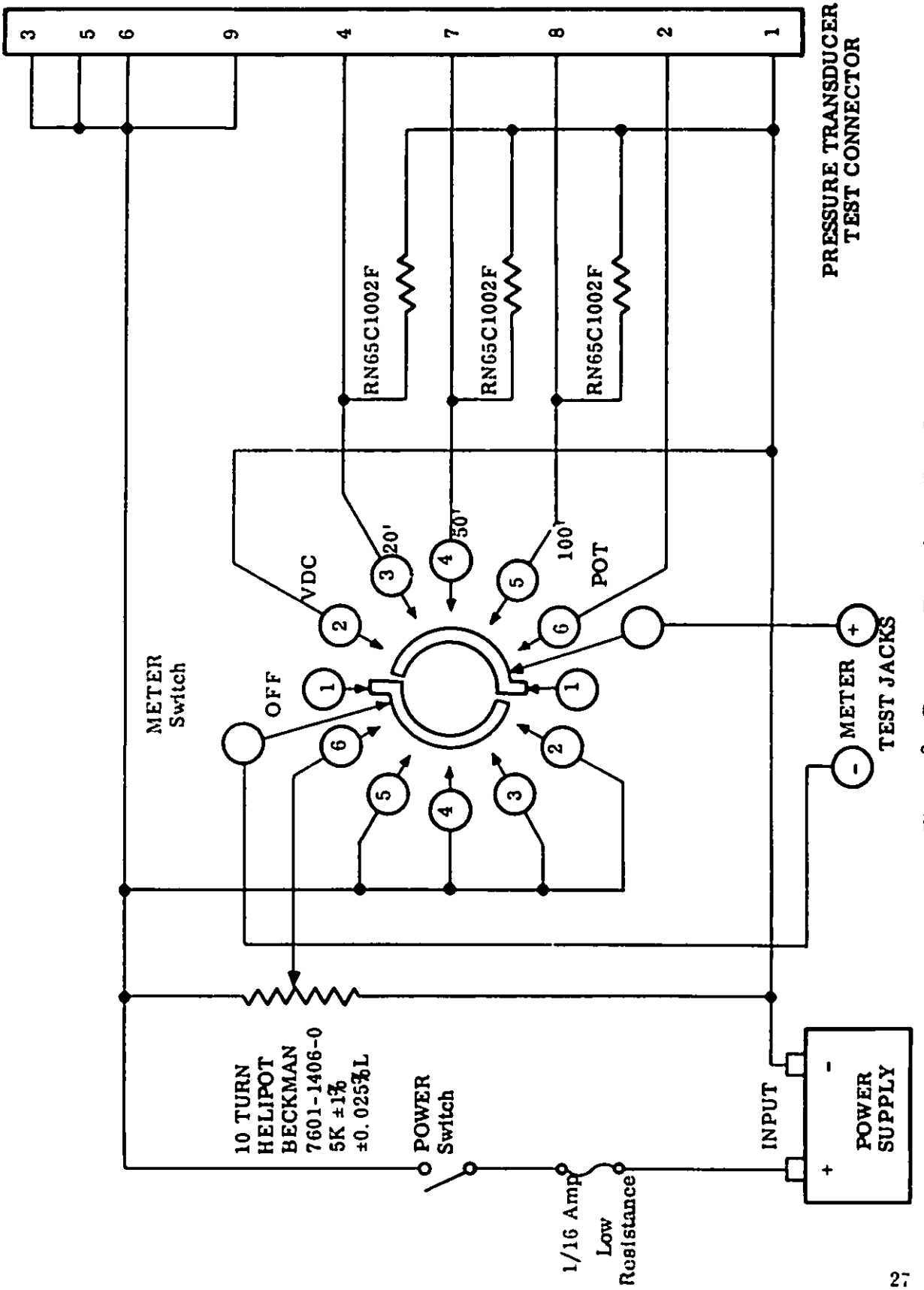


Figure 8. Pressure Transducer Test Set

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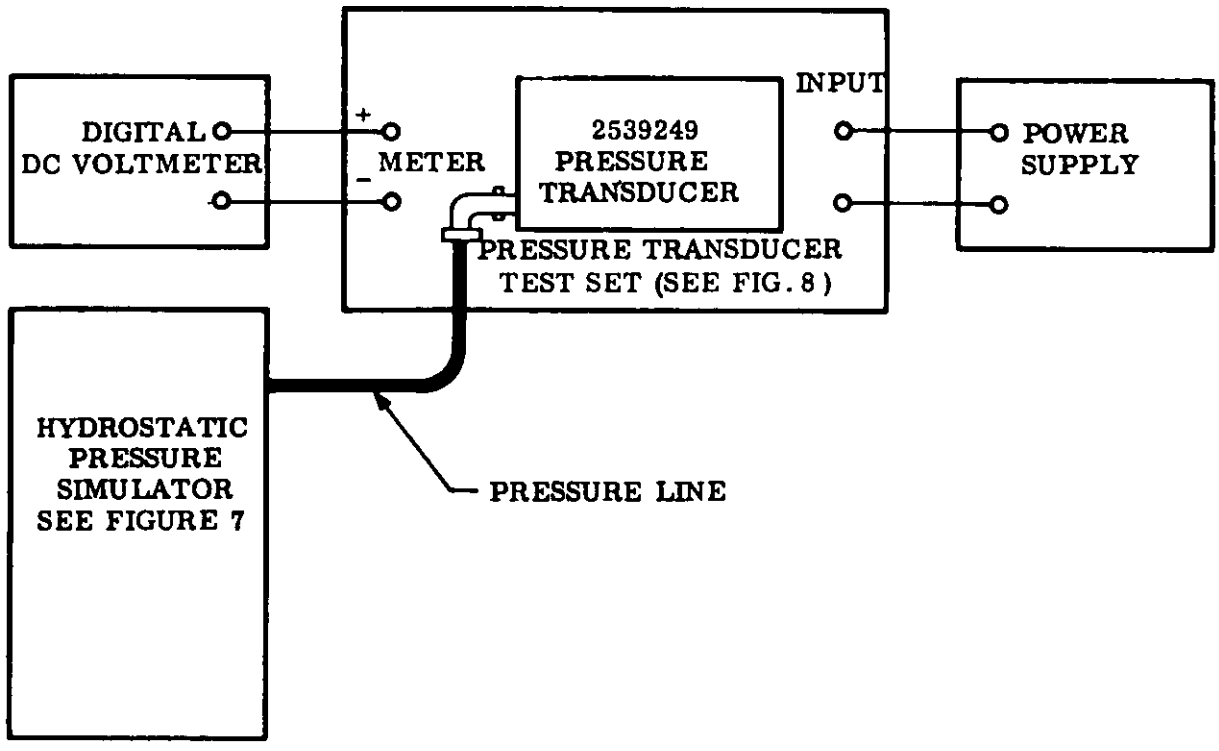


Figure 9. Pressure Transducer Test Setup

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4.7.3.3.8 Position the METER switch to 50'. The DVM shall indicate 29.6 VDC minimum.

4.7.3.3.9 Position the METER switch to 20'. The DVM shall indicate 29.6 VDC minimum.

NOTE

Increase HPS to indicated input pressure. Do not overshoot indicated pressure, and then decrease pressure to obtain proper indication. If the indicated pressures are exceeded on increasing pressure, return HPS to 0 psig and then increase pressure to obtain specified input. Use the 60 psi gage for pressure indications in 4.7.3.3.10 through 4.7.3.3.14 and 4.7.3.3.30 through 4.7.3.3.34; use the 250 psig gage in 4.7.3.3.15 through 4.7.3.3.18; use the 550 psig gage in 4.7.3.3.19 through 4.7.3.3.23 and 4.7.3.3.29; use 1000 psig gage in 4.7.3.3.24.

4.7.3.3.10 Slowly adjust the HPS regulator CW until the DVM first indicates less than 0.035 VDC. The HPS shall indicate 8.7 to 12.7 psig.

4.7.3.3.11 Position the METER switch to 50'. Slowly adjust HPS regulator CW until the DVM first indicates less than 0.035 VDC. The HPS shall indicate 22.0 to 26.0 psig.

4.7.3.3.12 Set the HPS to 0 psig. Increase the HPS for 25.0 ±0.2 psig. Position METER switch to POT. Adjust the null control dial for minimum indication of less than 0.003 vdc on the DVM (hereinafter called adjust for null). The control dial indication shall be 0.43 to 0.57.

4.7.3.3.13 Position the METER switch to 100'. Slowly adjust the HPS regulator CW until the DVM first indicates less than 0.035 VDC. The HPS shall indicate 42.5 to 46.5 psig.

4.7.3.3.14 Position the METER switch to POT. Adjust the HPS for 55.5 ±0.2 psig indication. Adjust the NULL. The control dial reading shall be 1.04 to 1.18.

4.7.3.3.15 Adjust the HPS for 75.0 ±0.5 psig. Readjust the HPS as accurately as possible for 55.5 psig (do not overshoot). Adjust for null and record null dial indication. The control dial shall indicate within ±0.05 of indication in paragraph 4.7.3.3.14. Record difference between 4.7.3.3.15 and 4.7.3.3.14.

4.7.3.3.16 Adjust the HPS for 95.0 ±0.5 psig indication. Adjust for null. The NULL control dial reading shall be 1.81 to 1.99.

4.7.3.3.17 Adjust the HPS as accurately as possible for 120.0 ±0.5 psig indication. Adjust for null. The NULL control dial reading shall be 2.31 to 2.49.

4.7.3.3.18 Adjust the HPS for 220.0 ±0.5 psig indication. Adjust for null. The NULL control dial reading shall be 4.30 to 4.50.

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4.7.3.3.19 Adjust the HPS for 300.0 ±0.5 psig indication. Adjust for null. The NULL control dial reading shall be 5.88 to 6.12.

4.7.3.3.20 Adjust the HPS for 330.0 ±0.5 psig indication. Adjust for null. The NULL control dial reading shall be 6.48 to 6.72.

4.7.3.3.21 Readjust the HPS as accurately as possible for 300.0 psig indication (do not overshoot). Adjust for null and record NULL control dial indication. The control dial shall indicate within ±0.05 of indication in 4.7.3.3.19. Record difference between 4.7.3.3.19 and 4.7.3.3.21.

4.7.3.3.22 Adjust theHPS for 440.0 ±0.5 psig. Adjust for null. The NULL control dial shall indicate 8.67 to 8.93.

4.7.3.3.23 Adjust the NULL control dial to 10.00. Adjust the HPS until the DVM no longer indicates a change in voltage greater than 0.003 vdc as pressure is increased. The indicated pressure shall be 500 ±6 psig.

4.7.3.3.24 Adjust the HPS to 750 +0, -25 psig. Adjust for NULL. The NULL control dial reading shall be 10.00 ±0.10.

4.7.3.3.25 Turn the HPS Leak Check cut-off valve maximum CW. Rotate the regulator knob one turn CCW. The indicated pressure shall not change more than 1 psig in 5 minutes.

NOTE

If leak is detected, recheck all pressure fittings to assure tightness and repeat test.

4.7.3.3.26 Position the METER switch to 20'. The DVM shall indicate 0.035 VDC maximum.

4.7.3.3.27 Position the METER switch to 50'. The DVM shall indicate 0.035 VDC maximum.

4.7.3.3.28 Position the METER switch to 100'. The DVM shall indicate 0.035 VDC maximum.

4.7.3.3.29 Position the METER switch to POT. Turn the HPS Leak Check cut-off valve maximum CCW. Slowly decrease the HPS pressure until the DVM first indicates a change greater than 0.05 VDC. The HPS indication shall be 500 ±6 psig.

4.7.3.3.30 Adjust the NULL control dial to exactly 1.11. Adjust the HPS for null. Record the HPS indication.

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4.7.3.3.31 Position the METER switch to 100'. Slowly decrease the HPS pressure until the DVM indication is 29.6 VDC minimum. The indicated pressure shall be 42.5 to 46.5 psig. The separation between the indication in 4.7.3.3.30 and 4.7.3.3.31 shall be a minimum of 9.0 psi.

4.7.3.3.32 Position the METER switch to 50'. Slowly decrease the HPS pressure until the DVM indicates 29.6 VDC minimum. The indicated pressure shall be 22.0 to 26.0 psig.

4.7.3.3.33 Position the METER switch to 20'. Slowly decrease the HPS pressure until the DVM indicates 29.6 VDC minimum. The indicated pressure shall be 8.7 to 12.7 psig.

4.7.3.3.34 Slowly decrease the pressure to 0.0 psig. Position the METER switch to POT. Adjust for null. The NULL control dial reading shall be -0.01 to +0.06.

4.7.3.3.35 Position all POWER switches to OFF. Disconnect the HPS from the pressure transducer.

4.7.4 Visual inspections.

4.7.4.1 Identification marking. (Major)

4.7.4.1.1 Nomenclature, control callout and identification data shall be correct, clear and legible. (Reference: BuWeps 2539249.)

4.7.4.2 Interchangeability. (Major)

4.7.4.2.1 Terminals shall be correct. (Reference: BuWeps 2539249.)

4.7.4.2.2 Length shall be 2.015 inches maximum. (Reference: BuWeps 2539249.)

4.7.4.2.3 Width shall be 2.025 inches maximum. (Reference: BuWeps 2539249.)

4.7.4.2.4 Thickness shall be 1.435 inches maximum. (Reference: BuWeps 2539249.)

4.7.4.2.5 Length from center of elbow to mounting holes shall be 1:32 $^{+.03}_{-.06}$ inches maximum. (Reference: BuWeps 2539249.)

4.7.4.2.6 Elbow shall be 30° ±2° angle. (Reference: BuWeps 2539249.)

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4.7.5 Process verification.

4.7.5.1 Criteria stipulated in this paragraph and its subparagraphs are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.7.5.2 Lacing. (Minor)

4.7.5.2.1 Lacing and cable configuration shall be in accordance with procedures approved by the procuring activity. Specific method of verifying adequacy of cable lacing shall include visual and dimensional verification.

4.7.5.3 Soldering. (Major)

4.7.5.3.1 Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.7.5.4 Cleanliness. (Major)

4.7.5.4.1 Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.7.6 Packaging, packing and marking inspections.

4.7.6.1 Packaging. (Minor)

4.7.6.1.1 Verify that packaging is in accordance with Section 5.

4.7.6.2 Packing. (Major)

4.7.6.2.1 Verify that packing is in accordance with Section 5.

4.7.6.3 Marking. (Major)

4.7.6.3.1 Destination and other shipment markings shall be correct, legible and in accordance with Standard MIL-STD-129.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. See 6.2. Unit package shall consist of one valve.

5.1.1 Level A. Preservation and packaging of equipment for Level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished with Method D-1, or D-4. Preservation and packaging shall be method 1A.

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5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be accomplished in accordance with the supplier's own commercial practice when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified. See 6.2. Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with MIL-P-116, Method III.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The equipment, when preserved and packaged in accordance with 5.1.3, shall be overpacked in compliance with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect the valve from damage due to shock and vibration loads encountered during all modes of handling and transportation. Shock and vibration tests shall be in accordance with this specification.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The Pressure Transducer is intended for use in a torpedo.

6.1.2 Description. The Pressure Transducer is a mechanical assembly installed in the Automatic Pilot assembly of the Mark 46 torpedo.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Required data.

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(e) Assigned activity for preproduction and periodic tests.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification, the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Nonoperating. The condition of the uninstalled Pressure Transducer when stored, transported, or handled.

6.3.1.2 Ready, nonoperating. The condition of Pressure Transducer when assembled with end item, transported and handled.

6.3.1.3 Ready. The condition of the Pressure Transducer when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.1.4 Operating. Operating is the condition of the Pressure Transducer when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement of acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.3 Shock. Shock is an abrupt change in the position of a body, resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and in the form of a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body, resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature. The temperature of the environment to which the Pressure Transducer may be exposed, expressed in degrees Fahrenheit (° F).

6.3.6 Pressure. The pressure of the environment to which the Pressure Transducer may be exposed, expressed in pounds per square inch gage (psig).

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6.3.7 Humidity. The relative humidity of the environment to which the Pressure Transducer is exposed, expressed as a percent.

6.3.8 Switch closure. The condition when contact resistance does not exceed 10.0 ohms at 3.0 ±0.2 ma DC.

6.3.9 Switch open. The condition when the leakage between contacts of the switch does not exceed 20 ua at 30 ±2 VDC.

6.3.10 Burst. A rupture or deformation of the unit resulting in the operating medium transferring from the pressure chamber to the outer environment.

6.3.11 Energized. The application of the pressure medium to the pressure fitting and electrical signals to the potentiometer and switches.

6.3.12 Full pressure cycle. The application of a pressure medium resulting in a pressure change from zero (0) psig to 745 ±5 psig and then to zero (0) psig.

6.3.13 Switch chatter. The condition when a closed switch will momentarily open and then close repeatedly or an open switch will momentarily close and then open repeatedly. Chatter shall be defined as existing if the 35 mv closed voltage exceeds 2 volts or the 29.6 vdc open voltage drops to 27.6 vdc.

6.3.14 Equivalent noise resistance. The total residual noise voltage from terminal 2 to terminal 3 divided by the source current.

6.3.15 Potentiometer incremental pressure cycle. The application of a pressure medium resulting in a small pressure change and then reduction in pressure to the starting point at a specified rate and for a specified time. This may be at any average pressure and may be a change in pressure of 1 psi to 20 psi peak-to-peak.

6.3.16 Switch incremental pressure cycle. The application of a pressure medium resulting in the switch closing and opening at a specified rate and for a specified time.

6.3.17 Isolated circuit. Circuits existing between the following terminals:

(a) 1, 2 and 3

(b) 4 and 5

(c) 6 and 7

(d) 8 and 9

6.3.18 Defective unit. A unit shall be considered defective if it does not meet all examinations or tests specified, or deviations are found through visual examination after environmental exposure.

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6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A	This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.
Level B	Not applicable.
Level C	This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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REPORT FORM

QUALITY ASSURANCE PROVISIONS

ACCEPTANCE TESTS AND EXAMINATIONS

FOR

PRESSURE TRANSDUCER

2539249

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TESTS AND EXAMINATIONS

PRESSURE TRANSDUCER 2539249

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the acceptance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the pressure transducer bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The acceptance tests and examinations of 4.7 shall be compiled and the results recorded or verified in this report form prior to acceptance of the pressure transducer. Unless otherwise noted, the sequence of testing, recording and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

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30. ACCEPTANCE TESTS AND EXAMINATIONS

		Reference	Limits	Reading
30.1	Test equipment	4.7.2	In calibration	_____
30.2	Electrical tests			
30.2.1	Initial tests			
	DC resistance	4.7.3.1.1	4700 to 5300 ohms at 79°F	_____
	Resistance: Case to each terminal	4.7.3.1.2	1 megohm min	_____
	Resistance: Terminal 1 to 4, 5, 6, 7, 8 and 9		1 megohm min	_____
	Terminal 4 to 6, 7 8 and 9		1 megohm min	_____
	Terminal 6 to 8 and 9		1 megohm min	_____
30.3	Noise tests			
30.3.1		4.7.3.2.3	29.9 to 30.1 vdc	_____
30.3.2	Up-Down axis	4.7.3.2.6	300 ohm noise max	_____
30.3.3	Transverse axis	4.7.3.2.7	300 ohm noise max	_____
	Fwd-aft axis		300 ohm noise max	_____
30.4	Operational tests			
30.4.1	30 VDC Power Supply	4.7.3.3.5	29.9 to 30.1 VDC	_____
30.4.2	DVM indication (POT)	4.7.3.3.6	0.175 VDC max	_____
30.4.3	DVM indication (100')	4.7.3.3.7	29.6 VDC min	_____
30.4.4	DVM indication (50')	4.7.3.3.8	29.6 VDC min	_____
30.4.5	DVM indication (20')	4.7.3.3.9	29.6 VDC min	_____
30.4.6	20' switch	4.7.3.3.10	8.7 to 12.7 psig	_____
30.4.7	50' switch	4.7.3.3.11	22.0 to 26.0 psig	_____

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		Reference	Limits	Reading
30.4.8	25.0 ±0.2 psig	4.7.3.3.12	0.43 to 0.57	_____
30.4.9	100' switch	4.7.3.3.13	42.5 to 46.5 psig	_____
30.4.10	55.5 ±0.2 psig	4.7.3.3.14	1.04 to 1.18	_____
30.4.11	Null	4.7.3.3.15	NULL dial indication	_____
	Repeatibility	4.7.3.3.15	Within 0.05 of 4.7.3.3.14	_____
30.4.12	95.0 ±0.5 psig	4.7.3.3.16	1.81 to 1.99	_____
30.4.13	120.0 ±0.5 psig	4.7.3.3.17	2.31 to 2.49	_____
30.4.14	220.0 ±0.5 psig	4.7.3.3.18	4.30 to 4.50	_____
30.4.15	300.0 ±0.5 psig	4.7.3.3.19	5.88 to 6.12	_____
30.4.16	330.0 ±0.5 psig	4.7.3.3.20	6.48 to 6.72	_____
30.4.17	Null	4.7.3.3.21	Null dial indication	_____
	Repeatability	4.7.3.3.21	Within ±0.05 of 4.7.3.3.19	_____
30.4.18	440.0 ±0.5 psig	4.7.3.3.22	8.67 to 8.93	_____
30.4.19	Active range	4.7.3.3.23	494 to 506 psig	_____
30.4.20	750 $^{+0}_{-25}$ psig	4.7.3.3.24	9.90 to 10.10	_____
30.4.21	No detectable leak	4.7.3.3.25	1 psi max in 5 min	_____
30.4.22	DVM indication (20')	4.7.3.3.26	0.035 VDC max	_____
30.4.23	DVM indication (50')	4.7.3.3.27	0.035 VDC max	_____
30.4.24	DVM indication (100')	4.7.3.3.28	0.035 VDC max	_____
30.4.25	Pot end	4.7.3.3.29	494 to 506 psig	_____
30.4.26	HPS null for 1.11 setting	4.7.3.3.30	Record HPS reading	_____

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		Reference	Limits	Reading
30.4.27	100' switch	4.7.3.3.31	42.5 to 46.5 psig	_____
			Separation 9.0 psi min	_____
30.4.28	50' switch	4.7.3.3.32	22.0 to 26.0 psig	_____
30.4.29	20' switch	4.7.3.3.33	8.7 to 12.7 psig	_____
30.4.30	0 psig	4.7.3.3.34	-0.01 to +0.06	_____
30.5	Visual inspection			
30.5.1	Identification marking	4.7.4.1		_____
30.5.2	Interchangeability			
	Terminals	4.7.4.2.1		_____
	Length	4.7.4.2.2	2.015 inches max	_____
	Width	4.7.4.2.3	2.025 inches max	_____
	Thickness	4.7.4.2.4	1.435 inches max	_____
	Length (center of elbow to mounting holes)	4.7.2.5	1.26 to 1.35 in.	_____
	Elbow angle	4.7.4.2.6	28° to 32°	_____
30.6	Packaging, packing and marking inspections			_____
30.6.1	Packaging	4.7.6.1		_____
30.6.2	Marking	4.7.6.3		_____

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-P-82459(OS) PRESSURE TRANSDUCER (FOR TORPEDO MK 46 MOD 0 and MOD 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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